From OAM Multiplexing through Quantum Key Distribution to Photonic Transceiver Tolerance

Worked Examples in Communication

Version history:
- **v2 (Mar 23, 2026):** Full document aligned with D1 (C1 v12) template. Five worked examples with Hamiltonians, derivation roadmaps, experimental evidence, bridge synthesis, and consolidated evidence table.

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Abstract

Every optical communication system — from OAM multiplexers to QKD ground stations to silicon photonic transceivers — is designed using classical simulation tools (Code V, Zemax, RSoft, LightTools). Each tool is effective within its domain, but each embodies a physical model with hard boundaries. Code V does not know that its Strehl ratio is the quantum fidelity, or that quantum-grade optics requires tolerances 45× tighter than the classical Maréchal criterion. RSoft does not cascade $N(N-1)/2$ MZIs with correlated errors to predict system yield. No commercial tool computes the OAM configuration entropy that collapses a full crosstalk matrix into a single quality-control number, or the parity cancellation advantage that quantifies when entangled photons outperform classical light by orders of magnitude.

This preliminary document presents five end-to-end worked examples that identify exactly where each tool stops and what physics lies beyond. The examples form a complexity staircase: OAM multiplexing (WE-C-1), QKD link budget (WE-C-2), free-space link through atmospheric turbulence (WE-C-3), silicon photonic transceiver tolerance (WE-C-4), and boson sampling processor fidelity (WE-C-5). A master communication-link propagator, $K_{\text{link}} = K_{\text{det}} \cdot K_{\text{channel}} \cdot K_{\text{source}}$, serves as the unifying equation: each worked example begins with an explicit Hamiltonian, derives the corresponding propagator factor, and evaluates it on both classical (coherent-state) and quantum (Fock-state or entangled) inputs through a three-stage structure. Three bridge equations connect classical optical figures of merit to quantum communication metrics, and a consolidated table of 20 experiments (1984–2025) grounds the framework in measured physics.

Three results appear across the five examples. First, the configuration entropy from the B1 singularity framework is reinterpreted as an engineering security classifier for high-dimensional OAM-based QKD. Second, the entanglement parity cancellation advantage is computed quantitatively for Kolmogorov turbulence: $A \approx 8.7$ at $W_{\text{rms}} = \lambda/10$ and $A \approx 5700$ at $W_{\text{rms}} = \lambda/5$, showing that entangled photon pairs substantially outperform separable states under atmospheric aberrations dominated by odd-parity Zernike modes. Third, the tolerance–hardness duality for boson sampling processors is identified: the fabrication regimes where quantum advantage is hardest to simulate classically are precisely the regimes where the tolerance window is narrowest, defining a practical design-space boundary for photonic quantum advantage demonstrations.

This preliminary technical document aims to bridge classical and quantum optical en gineering, enabling the seamless deployment of a unified quantum photonic engineering architecture, with worked examples in communication. It is in the series D, worked-example database. The first document of the series D is D1: "From Fresnel to Fock Space: Bridging Classical and Quantum Optical Engineering - Worked Examples with Hamiltonian Framework."

List of Abbreviations and Symbols

Optical Engineering Tools			
ABCD	Ray transfer matrix	AO	Adaptive optics
BPM	Beam propagation method (RSoft)	Code V	Ray-tracing lens design (Synopsys)
LightTools	Monte Carlo illumination (Synopsys)	RSoft	Photonic simulation (Synopsys)
SLM	Spatial light modulator		
Quantum Optics and Photonics			
BB84	Prepare-and-measure QKD protocol	BBM92	Entanglement-based QKD protocol
BS	Beam splitter	HOM	Hong–Ou–Mandel effect
NOON	$\|N,0\rangle + \|0,N\rangle$ state	OAM	Orbital angular momentum
PUF	Physical unclonable function	QBER	Quantum bit error rate
QKD	Quantum key distribution	SKR	Secure key rate
SNSPD	Supercond. nanowire detector	SPDC	Spontaneous parametric down-conversion
TVD	Total variation distance		
Framework-Specific Terms			
B1–B6	Six-pillar report series	C1–C7	Production-grade toolkit series
D1–D6	Worked-example database	DR4–DR6	Design rules (quantum-enhanced)
WE-C-k	Worked Example k (Comm.)		

Notation Convention

Two quantities in D6 share notation that can cause confusion:

- S = Strehl ratio (classical wavefront quality, WE-C-2/C-3). Defined as $S = \exp[-(2\pi W_{\mathrm{rms}}/\lambda)^2]$.
- S = configuration entropy (OAM quality metric, WE-C-1). Defined as $S = -\sum_j p_j \ln p_j$.

Context always disambiguates: Strehl appears with W_{rms}; entropy appears with the crosstalk matrix M. Where both appear in the same paragraph, we write S_{Strehl} and S_{ent} explicitly.

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1 Introduction

1.1 From OAM to System Yield: What the Title Means

This document, D6, traces the life of a photon in a communication system from source to detection, through five progressively more complex examples:

WE-C-1 OAM source	$\longrightarrow$	**WE-C-2** quantum channel (QKD)	$\longrightarrow$	**WE-C-3** turbulent atmosphere		
		$\longrightarrow$	**WE-C-4** photonic transceiver	$\longrightarrow$	**WE-C-5** boson processor	

This parallels D1's "Fresnel to Fock Space" journey but is organized by application domain (communication) rather than by formalism (Hamiltonian). The real motivation: an engineer designing a telecom OAM multiplexer, a QKD ground station, or a photonic transceiver uses Code V, RSoft, or LightTools — and each tool has a hard boundary where it silently stops being valid. However, here D6 shows exactly where those boundaries are and what physics lives on the other side. That is the practical value proposition.

1.2 The Unifying Equation: Communication-Link Propagator

In D1, the master Hamiltonian (D1 Eq. 10) and its product-of-unitaries form (D1 Eq. 11) serve as the Rosetta Stone connecting all seven worked examples. D6 requires an analogous unifying equation specific to communication systems.

Why a propagator, not a transfer matrix or S-parameter? The practicing engineer's instinct is to use transfer matrices (for free-space links) or S-parameters (for photonic circuits). These are the natural languages of Code V and RSoft, respectively. However, both formalisms carry a hidden limitation: they describe *field amplitudes* (classical) but not *photon-number states* (quantum). A transfer matrix maps $E_{\text{out}} = T \cdot E_{\text{in}}$; it cannot describe the Hong–Ou–Mandel effect, entanglement generation at a beam splitter, or sub-Poissonian photon statistics — all of which are essential for WE-C-2 (QKD) and WE-C-5 (boson sampling).

The propagator K resolves this: it maps *any* input state — coherent, Fock, squeezed, or entangled — to the corresponding output state through a single object. When the input is a coherent state $|\alpha\rangle$, K reduces to the classical transfer matrix (Stage 1). When the input is a Fock state $|n\rangle$, K produces the full quantum output including non-classical correlations (Stage 3). The three-stage classical-to-quantum structure (Section 1.3) is therefore not imposed from outside — it emerges naturally from the propagator evaluated on different input states.

Every communication link, classical or quantum, is described by a three-factor propagator:

$$\boxed{K_{\text{link}} = K_{\text{det}}(\eta_{\text{det}}, R_{\text{dark}}) \;\cdot\; K_{\text{channel}}(M, \sigma_\phi, C_n^2) \;\cdot\; K_{\text{source}}(\ell, \mu, |\psi_{\text{in}}\rangle)} \tag{1}$$

where:

- K_{source} generates the optical state: OAM mode ℓ, mean photon number μ, or entangled state $|\psi_{\text{in}}\rangle$.
- K_{channel} propagates through the medium: crosstalk matrix M (WE-C-1), fiber loss (WE-C-2), turbulence phase ϕ (WE-C-3), MZI mesh unitary U (WE-C-4/C-5).
- K_{det} models the receiver: detector efficiency η_{det}, dark count rate R_{dark}.

Connection to D1's master Hamiltonian. D1's product-of-unitaries $U_{\text{total}} = U_4 U_3 U_2 U_1$ (D1 Eq. 11) describes a photon traversing four types of interaction inside a single device. D6's propagator product $K_{\text{link}} = K_{\text{det}} \cdot K_{\text{channel}} \cdot K_{\text{source}}$ describes a photon traversing three system-level stages in a communication link. The mathematical structure is the same (composition of linear maps); the physical granularity is different (device-level in D1, system-level in D6). Each K factor in D6 may itself be a product of D1-style unitaries: for example, K_{channel} in WE-C-4 is $\prod_{k=1}^{M} U_k(\theta_k, \phi_k)$ for the MZI mesh.

Strengths and limitations of the propagator approach.

Table 1: Strengths and limitations of the propagator formalism for communication systems.

Strengths	
Unified classical–quantum	Same K handles coherent, Fock, squeezed, and entangled inputs; no formalism switch between Stages 1–3.
Natural composition	$K_{\text{link}} = K_3 \cdot K_2 \cdot K_1$; system-level analysis = product of subsystem propagators.
Path-integral connection	K is the kernel of the Feynman path integral; tolerance analysis via B6 (disorder-averaged propagator) follows directly.
Blind-spot identification	The $K_i \to K_{i+1}$ interface is where classical tools lose track of the quantum state — the propagator makes this explicit.
Limitations	
Not directly callable	No commercial tool (Code V, RSoft, LightTools) computes K in its quantum form; custom Python code is required.
Mode-basis dependence	K is defined in a specific mode basis (LG for OAM, waveguide modes for PIC); changing basis requires explicit transformation.
Scalability	For N-mode systems, K is an $N \times N$ matrix (classical) or an operator on N-mode Fock space (quantum); the quantum case grows exponentially with photon number.
Open-system extension	Loss and decoherence require the Feynman–Vernon influence functional (B6 Bridge 5), which is more complex than the simple product form of Eq. (1).

Central Theme

The master propagator Eq. (1) is the D6 Rosetta Stone. Each classical tool computes one factor: Code V computes K_{channel} for free-space propagation; RSoft computes K_{channel} for guided-wave devices; LightTools computes stray-light contributions to K_{det}. No classical tool computes the full product, and none tracks the quantum state across the $K_{\text{source}} \to K_{\text{channel}}$ interface. The blind spots arise precisely at the interfaces between factors — just as in D1's product-of-unitaries.

The five worked examples each instantiate Eq. (1) with different physics for each factor:

Table 2: How each WE-C instantiates the master propagator.

WE	K_{source}	K_{channel}	K_{det}
C-1	OAM mode u_ℓ via spiral phase plate	Crosstalk matrix M_{jk} under misalignment	Bucket detector (power in mode ℓ)
C-2	Entangled pairs (SPDC) or WCP $\lvert\alpha\rangle$	Fiber loss $10^{-\alpha L/10}$; free-space η_{FSO}	SNSPD ($\eta = 0.90$, $R_{\text{dark}} = 1$ Hz)
C-3	Gaussian beam or OAM mode	Kolmogorov phase screen $e^{i\phi(\mathbf{r})}$	Coherent receiver (homodyne/heterodyne)
C-4	Laser ($\lvert\alpha\rangle$) or single photon	MZI mesh $U = \prod_k U_k(\theta_k, \phi_k)$	Photodetector array
C-5	n-photon Fock state $\lvert 1\rangle^{\otimes n}$	Haar-random unitary (boson sampling)	Photon-number-resolving detector

1.3 The Three-Stage Classical-to-Quantum Structure

Each worked example follows the three-stage structure from B6:

1. **Stage 1 (Classical):** K_{link} evaluated for coherent states; classical FOMs (BER, capacity, Strehl).
2. **Stage 2 (Hamiltonian/CMT):** Transfer-matrix or coupled-mode description; bridge equations identified.
3. **Stage 3 (Quantum):** K_{link} evaluated for Fock states or entangled states; quantum FOMs (QBER, SKR, fidelity, TVD).

The classical-to-quantum boundary lives in K_{source}, not in K_{channel}.

1.4 What Is New in This Report

Table 3: Content comparison: D1 (C1 v12) vs. this D6 report.

Content	**D1**	**D6**
Master equation	Hamiltonian Eq. 10–11	Communication propagator Eq. (1)
Organizing axis	Formalism (Hamiltonian)	Application (communication)
Examples	7 (single devices)	5 (end-to-end systems)
OAM entropy as capacity metric	—	**New**
QKD link budget with $S = F$	—	**New**
Cancellation advantage A (Kolmogorov)	Eq. 54 in B4	Quantitative link budget
Turbulence-as-PUF entropy	—	**Speculative (new)**
Tolerance–hardness duality	—	**New**
Bridge equations	3 (B3–B4–B5)	3 (adapted to communication)
Experimental evidence table	27 experiments	20 experiments (communication-specific)

1.5 Document Structure

Section 2 presents the master table. Sections 3–7 develop all five worked examples following the D1 per-WE template: Hamiltonian, derivation roadmap, simulation outputs, what is not shown, what should be highlighted, experimental evidence, classical–quantum connection, and honest assessment. Section 8 synthesizes the three bridge equations. Section 9 provides the consolidated experimental evidence table. Section 11 concludes.

2 Master Table

Table 4 organizes all five worked examples along eight columns, following the format of D1 Table 2: the Hamiltonian that governs the physics, the classical tool that partially covers it, the Python simulation that fills the gap, the pillar assignment within the B1–B6 framework, the blind spot that no classical tool can access, and the key figure of merit. The "Blind Spot" column is the most important: it identifies, for each example, the specific physics that the engineer's existing tools cannot compute. This is where D6 adds value.

Table 4: Master table: five worked examples in communication (cf. D1 Table 2).

WE	Hamiltonian	Classical Tool	Python Sim.	Pillar	Blind Spot	Key FOM	TRL
C-1	Free OAM modes + displacement	Code V: far-field OAM overlap	`d6_we_c1`	B1, B3	Entropy S as FOM; Class B recovery; QKD security link	Capacity (bits/s/Hz), S (nats)	3–5
C-2	QKD source–channel–detector chain	Code V: Strehl / tolerance	`d6_we_c2`	B4, B3	QBER, SKR, 45× tightening, cancellation A	SKR (bits/pulse), QBER	6–8
C-3	Free propagation + turbulence potential	RSoft: BPM through index fluctuations	`d6_we_c3`	B5, B1	OAM purity degradation, PUF entropy, advantage A	BER, S_{turb}, r_0	4–6
C-4	MZI mesh (product of unitaries)	RSoft: per-device S-parameters	`d6_we_c4`	B6, B3	System-level F_{sys}, correlated errors, Maslov phase	Yield Y_{sys}, IL (dB)	4–5
C-5	Haar-random unitary (boson sampling)	None (classical intractable)	`d6_we_c5`	B5, B4, B6	TVD (not S^N), tolerance–hardness duality	F_U, TVD, β	2–3

3 WE-C-1: OAM Multiplexing

B1 Singularities **B3** Invariants **Master propagator:** $K_{\text{source}} = \text{OAM mode } u_\ell$

3.1 Hamiltonian

The free-space OAM mode set is described by N independent bosonic modes with topological charges $\ell_1, \ldots, \ell_N$:

$$H_{\text{WE-C1}} = \sum_{j=1}^{N} \hbar\omega \, a_{\ell_j}^\dagger a_{\ell_j}. \tag{2}$$

Under lateral misalignment s, the field transformation mixes modes via the unitary overlap matrix U_{jk}:

$$a'_{\ell_j} = \sum_k U_{jk} \, a_{\ell_k}, \quad \text{where} \quad M_{jk} = |U_{jk}|^2 \tag{3}$$

is the power crosstalk matrix (Eq. 6). The system specification is given in Table 5.

Table 5: WE-C-1 input specification.

Parameter	**Value**	**Unit**	**Notes**
Wavelength	1550	nm	Telecom C-band
Mode count N	4, 8, 16	—	Parametric sweep
Beam waist w_0	5	mm	At transmitter
s/w_0	0–0.5	—	30-point sweep
SNR	20	dB	Per mode

3.2 From Hamiltonian to Propagator

The free-mode Hamiltonian Eq. (2) generates trivial time evolution for aligned beams: each mode acquires phase $e^{-i\omega t}$ independently, and the propagator is diagonal. The non-trivial physics enters through the displacement s: the propagator for the misaligned channel is

$$K_{\text{channel}}^{(\text{C-1})} = U_{jk}(s) = \langle u_{\ell_j}(\mathbf{r}) \,|\, u_{\ell_k}(\mathbf{r} - s\,\hat{x})\rangle, \tag{4}$$

where U_{jk} is the unitary overlap matrix between centered and displaced LG modes. The power crosstalk matrix is $M_{jk} = |U_{jk}|^2$. In the language of the master propagator (Eq. 1): K_{source} generates OAM mode u_ℓ; $K_{\text{channel}} = U_{jk}(s)$ is the displacement-induced mixing; K_{det} is mode-resolved power detection (diagonal POVM).

When the input is a coherent state $|\alpha_\ell\rangle$ (Stage 1), K_{channel} acts as a classical transfer matrix on field amplitudes. When the input is a single photon $|1_\ell\rangle$ (Stage 3), the same U_{jk} determines the quantum state after the channel, and the entropy S acquires its QKD security interpretation.

3.3 Analytical Derivation

Derivation Roadmap

1. Laguerre–Gauss mode $u_\ell(r,\theta)$ as eigenstate of L_z.
2. Lateral displacement s: transform to displaced coordinates (r', θ').
3. Overlap integral $\to$ crosstalk matrix $M_{jk} = |\langle u_{\ell_j} | u'_{\ell_k}\rangle|^2$.
4. Eigenvalue decomposition of $M \to$ configuration entropy $S = -\sum p_j \ln p_j$.
5. Quadratic scaling law: $S \approx \gamma_N (s/w_0)^2$ (B1 Eq. 13).
6. MMSE equalization: effective SINR from eigenvalues of $M^H M$.
7. Channel capacity: $C_{\text{MMSE}} = \sum_j \log_2(1 + \text{SNR} \cdot \lambda_j)$.

Step 1. The Laguerre–Gauss mode of order $(p = 0, \ell)$:

$$u_\ell(r,\theta) = \frac{\sqrt{2/\pi}}{w_0} \left(\frac{r\sqrt{2}}{w_0}\right)^{|\ell|} \exp\!\left(-\frac{r^2}{w_0^2}\right) e^{i\ell\theta}. \tag{5}$$

Step 3. The crosstalk matrix under displacement s:

$$M_{jk} = \left| \int_0^\infty \int_0^{2\pi} u_{\ell_j}^*(r,\theta)\, u_{\ell_k}(r',\theta')\, r\, dr\, d\theta \right|^2. \tag{6}$$

Step 4. Configuration entropy (B1):

$$S = -\sum_j p_j \ln p_j, \qquad S \approx \gamma_N (s/w_0)^2. \tag{7}$$

Operating classes (B1 Section 7).

- **Class A** ($S < 0.15$, $\kappa \approx 1$): near-ideal isolation.
- **Class B** ($0.15 \leq S < 0.50$, $\kappa < 1000$): recoverable via MMSE.
- **Class C** ($S \geq 0.50$, $\kappa > 1000$): information lost.

Step 7. Channel capacity after MMSE:

$$C_{\mathrm{MMSE}} = \sum_{j=1}^{N} \log_2\bigl(1 + \mathrm{SNR} \cdot \lambda_j(M^H M)\bigr). \tag{8}$$

Stage 3: Quantum extension. For high-dimensional QKD using N OAM modes [6], the entropy S becomes a single-number security diagnostic (Table 6).

Table 6: B1 operating classes reinterpreted for QKD security.

Class	MUX meaning	QKD security meaning
A	Near-ideal isolation	Full secure key rate
B	Recoverable crosstalk	Reduced SKR; post-processing cost $\sim f(\kappa)$
C	Information lost	Insecure: Eve's information exceeds Alice–Bob

3.4 Simulation Outputs

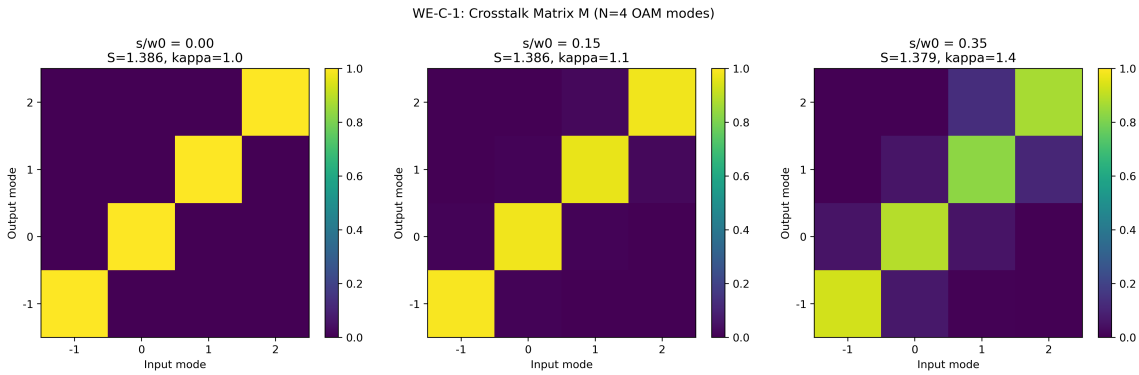

Figure 1: WE-C-1, Investigation 1: Crosstalk matrix M for $N = 4$ at $s/w_0 = 0$ (Class A), 0.15 (Class B), and 0.35 (approaching Class C).

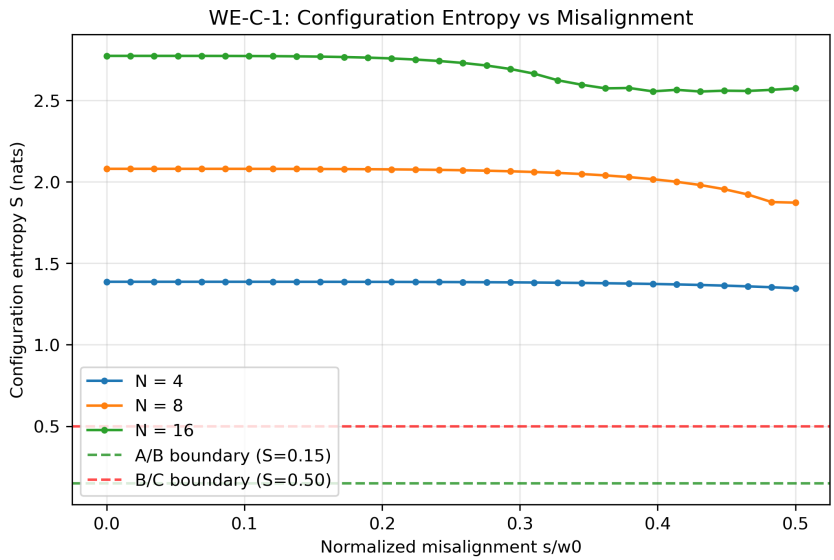

Figure 2: WE-C-1, Investigation 2: Entropy S vs. s/w_0 with A/B/C boundaries.

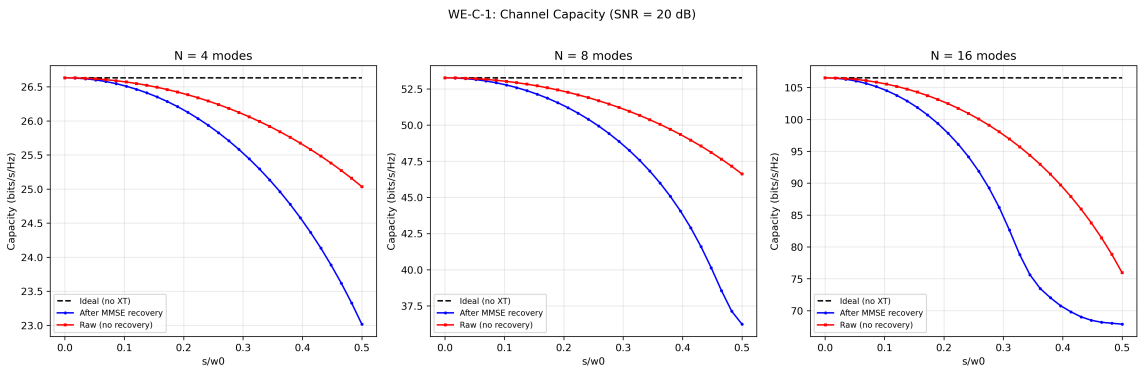

Figure 3: WE-C-1, Investigation 3: Capacity (raw, MMSE, ideal) vs. s/w_0. The gap between blue (MMSE) and red (raw) is the engineering value of Class B recovery.

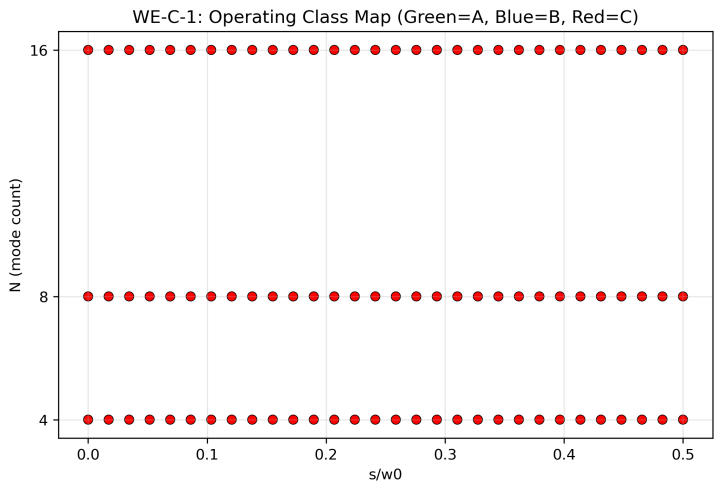

Figure 4: WE-C-1, Investigation 4: Operating class map (green = A, blue = B, red = C).

3.5 What Is Not Shown Here

1. Higher radial orders ($p > 0$) are omitted; they enlarge M and modify γ_N.
2. Atmospheric turbulence overlay is deferred to WE-C-3.
3. Finite receiver aperture truncation effects are not modeled.
4. Nonlinear crosstalk (high-power regime) is not treated.
5. Detector noise and finite-bandwidth effects are omitted.

3.6 What Should Be Highlighted

1. **Class B is the Feature Zone:** "defective" OAM lenses become functional multiplexers after computational recovery. The capacity gain ΔC quantifies this.
2. **The entropy S is a universal FOM:** it collapses the full $N \times N$ crosstalk matrix into a single number, enabling real-time quality monitoring.
3. **QKD security reinterpretation:** the B1 binning station doubles as a security monitor with no hardware change.

3.7 Experimental Evidence

Experimental Evidence (WE-C-1)

1. **Terabit OAM multiplexing** (Wang et al., 2012 [2]): 2.56 Tbit/s using 32 OAM states multiplexed with 42 wavelengths. Demonstrates that OAM multiplexing achieves Tbit/s-class capacity.
2. **OAM-MUX design considerations** (Xie et al., 2015 [4]): Crosstalk under lateral displacement quantified for FSO links.
3. **OAM spectrum correction** (Liu et al., 2008 [5]): Numerical correction for misalignment-induced OAM crosstalk — the "Class B recovery" concept in B1's language.
4. **High-dimensional QKD with OAM** (Mirhosseini et al., 2015 [6]): $d = 7$ OAM-based QKD achieving 2.05 bits/photon per sifted event.

Classical–Quantum Connection

What Code V / RSoft computes: Mode overlap integrals, coupling efficiency per OAM channel, far-field intensity.
How this emerges: $M_{jk} = |K_{\ell_j \ell_k}|^2$ in the LG basis.
Where the classical tool stops: No tool computes entropy S, performs A/B/C classification, or connects M to QKD security.

Honest Assessment

- Entropy S and Class B recovery: from B1 v35 (not new).
- New in D6: end-to-end capacity after MMSE; QKD security reinterpretation.
- Limitation: $p = 0$ only; no turbulence (deferred to WE-C-3).

4 WE-C-2: QKD Link Budget

B4 Coherence & Entanglement **B3** Invariants **Master propagator:** $K_{\text{source}} = \text{SPDC}$ entangled pairs

4.1 Hamiltonian

The QKD link decomposes into three sequential Hamiltonians:

$$H_{\text{WE-C2}} = \underbrace{H_{\text{SPDC}}}_{\text{source}} + \underbrace{H_{\text{fiber/FSO}}}_{\text{channel}} + \underbrace{H_{\text{SNSPD}}}_{\text{detector}}, \tag{9}$$

where $H_{\text{SPDC}} = i\hbar\chi^{(2)}(a_s^\dagger a_i^\dagger - a_s a_i)$ generates entangled signal–idler pairs, $H_{\text{fiber}} = \hbar\omega\, a^\dagger a$ with loss modeled as a beam-splitter coupling to a vacuum reservoir, and H_{SNSPD} models the detector response including efficiency η_{det} and dark counts.

Table 7: WE-C-2 input specification.

Parameter	**Value**	**Unit**	**Notes**
Source	SPDC (PPLN)	—	10^9 pairs/s
λ (fiber / FSO)	1550 / 810	nm	—
Fiber loss	0.2	dB/km	SMF-28
η_{det}	0.90	—	SNSPD
Dark count rate	1	Hz	—
f_{EC}	1.16	—	Cascade protocol
AO modes	0, 3, 10, 35	Zernike	Sweep

4.2 From Hamiltonian to Propagator

The three-term Hamiltonian Eq. (9) generates three sequential propagator factors:

$$K_{\text{link}}^{(\text{C-2})} = \underbrace{\Pi_{\text{SNSPD}}(\eta_{\text{det}})}_{K_{\text{det}}} \cdot \underbrace{B(\eta_{\text{ch}})}_{K_{\text{channel}}} \cdot \underbrace{S(r)}_{K_{\text{source}}}, \tag{10}$$

where:

- $S(r) = \exp[r(a_s^\dagger a_i^\dagger - a_s a_i)]$ is the two-mode squeezing operator generated by H_{SPDC}, producing entangled signal–idler pairs with squeeze parameter r.
- $B(\eta) = \exp[\theta(a^\dagger b - a b^\dagger)]$ is the beam-splitter operator that models channel loss: the signal mode a couples to a vacuum mode b with transmittance $\eta = \cos^2\theta$. For fiber: $\eta = 10^{-\alpha L/10}$. For free-space: $\eta = \eta_{\text{geo}} \cdot \eta_{\text{atm}} \cdot S_{\text{AO}}$.
- Π_{SNSPD} is the detector POVM with efficiency η_{det} and dark count probability p_{dark}.

The key connection to Bridge 2: when the channel is lossless ($\eta = 1$) but aberrated ($W_{\text{rms}} > 0$), the coupling into single-mode fiber is $\eta_{\text{coupling}} \approx S = F = \exp[-(2\pi W_{\text{rms}}/\lambda)^2]$. The Strehl ratio enters through K_{channel} as an effective beam-splitter transmittance.

4.3 Analytical Derivation

Derivation Roadmap

1. SPDC Hamiltonian $\to$ two-mode squeezed vacuum state $|\psi_{\text{SPDC}}\rangle$.
2. Channel loss as beam-splitter model: $a_{\text{out}} = \sqrt{\eta}\, a_{\text{in}} + \sqrt{1-\eta}\, a_{\text{vac}}$.
3. Fiber: $\eta_{\text{fiber}}(L) = 10^{-\alpha L/10}$. Free-space: $\eta_{\text{FSO}} = \eta_{\text{geo}} \cdot \eta_{\text{atm}} \cdot S_{\text{AO}}$.
4. Bridge 2 ($S = F$): coupling efficiency $\eta \approx S = \exp[-(2\pi W_{\text{rms}}/\lambda)^2]$.
5. QBER model: $(e_{\text{opt}}\, p_{\text{signal}} + \frac{1}{2} p_{\text{dark}})/(p_{\text{signal}} + p_{\text{dark}})$.
6. BB84 SKR: $R_{\text{sift}}[1 - H(e) - f_{\text{EC}}\, H(e)]$.
7. BBM92 SKR: coincidence rate $R_{\text{pair}}\, \eta_1\, \eta_2$; no parity cancellation.
8. Cancellation advantage: $A = \exp[2 f_{\text{odd}}(2\pi W_{\text{rms}}/\lambda)^2]$ with $f_{\text{odd}} \approx 0.87$.

Step 4: Bridge 2 ($S = F$). The Strehl ratio and quantum fidelity are identical:

$$S = F = \exp\left[-\left(\frac{2\pi W_{\text{rms}}}{\lambda}\right)^2\right]. \tag{11}$$

Bridge Equation

For QKD: the coupling into single-mode fiber is $\eta \approx S = F$, directly connecting the wavefront specification to quantum channel fidelity.
For NV-diamond sensing: $W_{\text{rms}} < \lambda/2800$, approximately 45× tighter than the classical Maréchal criterion $W_{\text{rms}} < \lambda/14$.

Step 6: BB84 secure key rate.

$$R_{\text{key}}^{\text{BB84}} = R_{\text{sift}}\left[1 - H(\text{QBER}) - f_{\text{EC}}\, H(\text{QBER})\right]. \tag{12}$$

Step 8: Cancellation advantage.

$$A = \exp\left[2\, f_{\text{odd}}\left(\frac{2\pi W_{\text{rms}}}{\lambda}\right)^2\right]. \tag{13}$$

For $W_{\text{rms}} = \lambda/10$: $A \approx 8.7$. For $W_{\text{rms}} = \lambda/5$: $A \approx 5700$.

Central Finding

The cancellation advantage grows *exponentially* with aberration strength. Atmospheric turbulence — dominated by odd-parity Zernike modes (tip, tilt, coma, $f_{\text{odd}} \approx 0.87$) — is precisely the regime where entangled pairs most outperform separable states.

4.4 Simulation Outputs

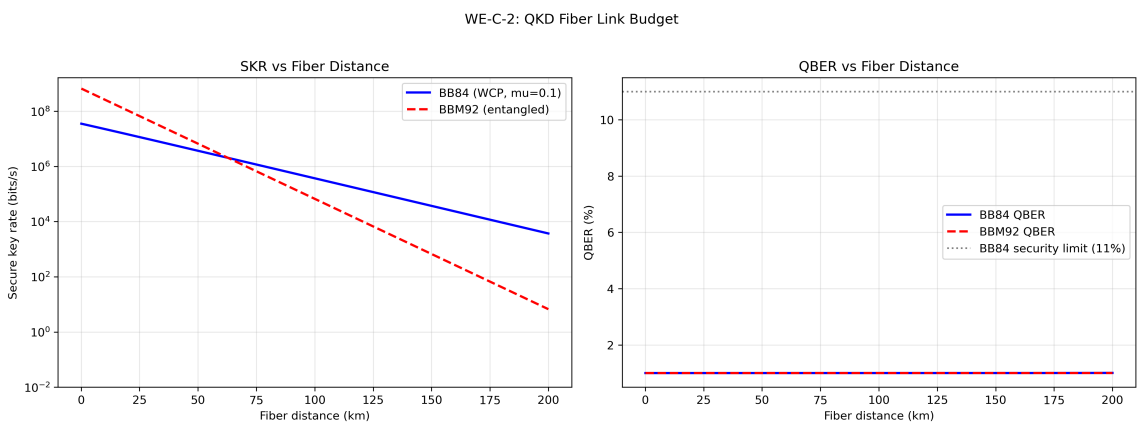

Figure 5: WE-C-2, Investigation 1: SKR and QBER vs. fiber distance (BB84 and BBM92).

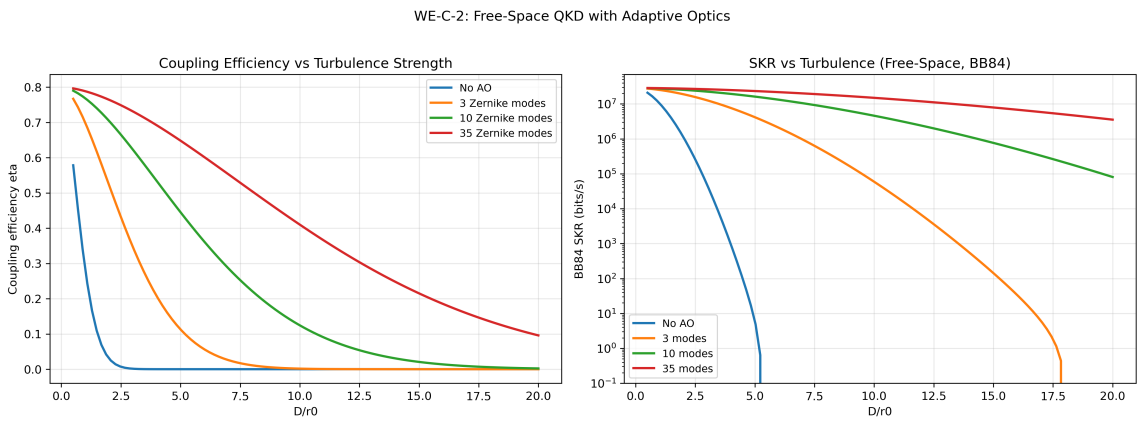

Figure 6: WE-C-2, Investigation 2: Coupling efficiency and SKR vs. D/r_0 with AO.

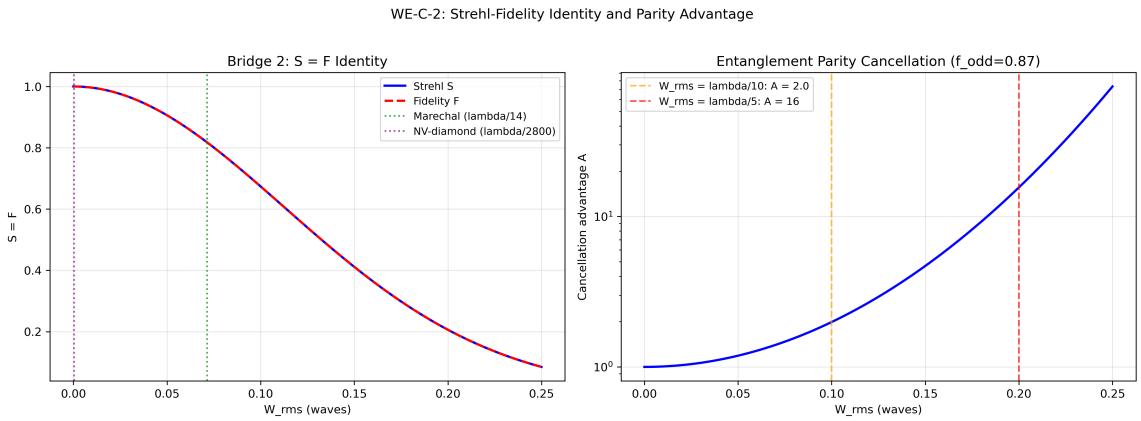

Figure 7: WE-C-2, Investigation 3: $S = F$ identity and cancellation advantage A.

4.5 What Is Not Shown Here

1. Finite-key effects (important for short satellite passes) require v2.
2. Decoy-state protocol for practical BB84 with WCP is not implemented.
3. Detector afterpulsing and timing jitter are modeled only through effective parameters.
4. Multi-mode fiber effects are not treated.
5. QKD over turbulent free-space with real-time AO feedback loop is deferred to WE-C-3.

4.6 What Should Be Highlighted

1. $S = F$ **in a link budget:** the optical engineer's Strehl ratio IS the QKD channel fidelity. This makes Code V's tolerance analysis directly relevant to QKD performance.
2. 45× **tightening:** quantum-grade optics requires $W_{\text{rms}} < \lambda/2800$, not $\lambda/14$.
3. **Cancellation advantage under Kolmogorov:** $A \approx 8.7$ at $W_{\text{rms}} = \lambda/10$ shows entanglement is not a marginal improvement — it is an order-of-magnitude advantage.
4. **No parity cancellation for dual-downlink:** BBM92 satellite QKD (Micius) cannot exploit the advantage because the two channels are independent.

4.7 Experimental Evidence

Experimental Evidence (WE-C-2)

1. **BB84 original** (Bennett & Brassard, 1984 [8]): foundational protocol.
2. **Commercial QKD** (ID Quantique, Toshiba, 2020s): fiber QKD at > 400 km.
3. **Satellite QKD — Micius** (Yin et al., 2017 [10]): entanglement-based QKD over 1200 km, demonstrating dual-downlink BBM92.
4. **AO-assisted satellite QKD** (arXiv:2411.09564, 2024 [11]): increasing SKR via adaptive optics on ground stations.
5. **Biphoton wavefront sensing** (Light: Sci. Appl., 2025 [12]): position-correlated biphoton Shack–Hartmann sensor validating the S_{12} framework.

Classical–Quantum Connection

What Code V computes: Strehl ratio, OPD tolerance, fringe visibility.
How this emerges: Code V's Strehl IS the quantum fidelity (Bridge 2).
Where Code V stops: It does not compute QBER, SKR, the 45× tightening, or the cancellation advantage A.

Honest Assessment

- BB84/BBM92 formulas: standard. New: $S = F$ in quantitative link budget; A for Kolmogorov.
- Limitation: asymptotic key rate; finite-key analysis needed for satellite passes.

5 WE-C-3: Free-Space Optical Link Through Turbulence

B5 Complexity/Chaos **B1** Singularities **Master propagator:** $K_{\text{channel}} = e^{i\phi(\mathbf{r})}$

5.1 Hamiltonian

Propagation through a turbulent atmosphere is described by:

$$H_{\text{WE-C3}} = \frac{p^2}{2k} + V_{\text{turb}}(\mathbf{r}, z), \tag{14}$$

where the turbulence potential $V_{\text{turb}}(\mathbf{r}, z) = k\,\delta n(\mathbf{r}, z)$ arises from refractive-index fluctuations δn with Kolmogorov statistics. The key parameter is the Fried parameter:

$$r_0 = \left[0.423\,k^2 \int_0^L C_n^2(z)\,dz\right]^{-3/5}. \tag{15}$$

5.2 From Hamiltonian to Propagator

The turbulence Hamiltonian Eq. (14) generates the propagator via the split-step method — which IS the discretized Feynman path integral:

$$K^{(\text{C-3})}_{\text{channel}}(\mathbf{r}_2, \mathbf{r}_1) = \underbrace{\mathcal{F}^{-1}\left[e^{-i|\mathbf{k}_\perp|^2\Delta z/2k}\right]}_{\text{free-space Fresnel step}} \cdot \underbrace{e^{i\phi(\mathbf{r})}}_{\text{phase screen}}, \tag{16}$$

where $\phi(\mathbf{r}) = k\int_0^{\Delta z} \delta n(\mathbf{r}, z')\,dz'$ is the accumulated turbulence phase. For a single thin screen (this document), the propagator is simply $K_{\text{channel}} = e^{i\phi(\mathbf{r})}$ applied to the input field. For

distributed turbulence (v2 extension), the full propagator is a product of N_{screen} split-step factors — the path-integral sum over all transverse positions at each screen.

The connection to RSoft is direct: RSoft's BPM computes exactly this split-step propagator for guided-wave structures. The turbulence phase screen replaces RSoft's refractive-index profile with a stochastic realization of δn. The physics is the same Huygens–Fresnel path integral; the input is different.

5.3 Analytical Derivation

Derivation Roadmap

1. Kolmogorov spectrum → von Kármán PSD → FFT phase screen generation.
2. Fried parameter r_0 from C_n^2 profile; D/r_0 ratio.
3. Noll residual variance $\sigma_\phi^2 = a_{N_Z}(D/r_0)^{5/3}$ after N_Z Zernike modes removed [7].
4. Strehl with AO: $S_{\text{AO}} = \exp(-\sigma_\phi^2)$.
5. Scintillation: $\sigma_I^2 = 1.23\, C_n^2\, k^{7/6}\, L^{11/6}$; log-normal intensity PDF.
6. BER averaged over scintillation for OOK/BPSK/16-QAM.
7. OAM mode purity: project $E_{\text{out}} = E_{\text{in}}\, e^{i\phi}$ onto LG basis (B1 connection).
8. Speculative: turbulence PUF entropy from transmission matrix (B5 connection).

Step 4: Strehl after AO.

$$S_{\text{AO}} = \exp\left(-a_{N_Z}\left(\frac{D}{r_0}\right)^{5/3}\right). \tag{17}$$

Step 5: Scintillation index.

$$\sigma_I^2 = 1.23\, C_n^2\, k^{7/6}\, L^{11/6}. \tag{18}$$

5.4 Simulation Outputs

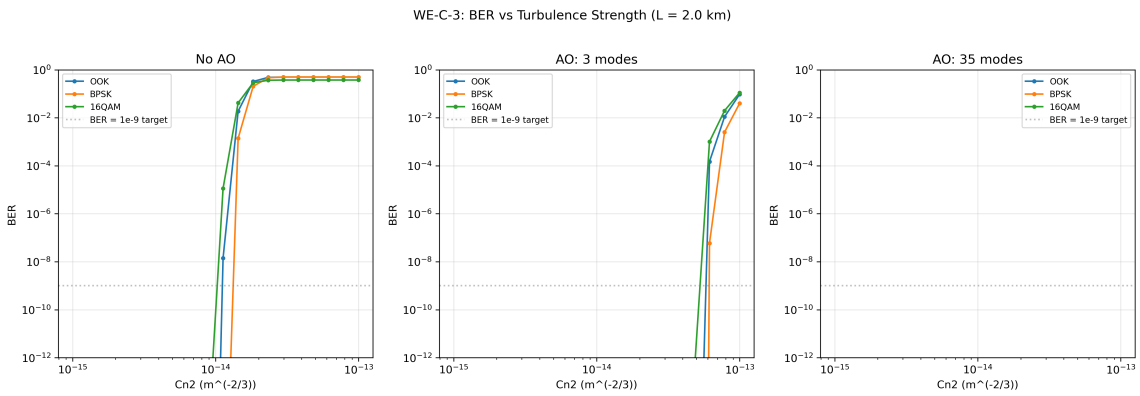

Figure 8: WE-C-3, Investigation 1: BER vs. C_n^2 for OOK/BPSK/16-QAM with AO.

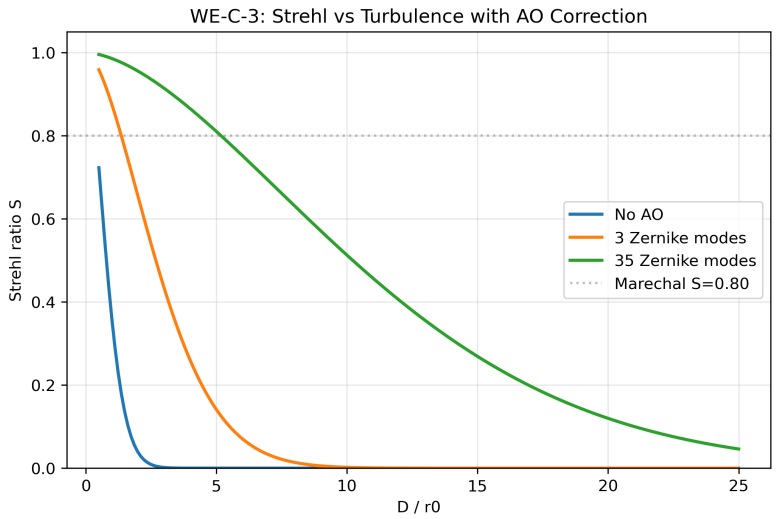

Figure 9: WE-C-3, Investigation 2: Strehl vs. D/r_0 with AO correction levels.

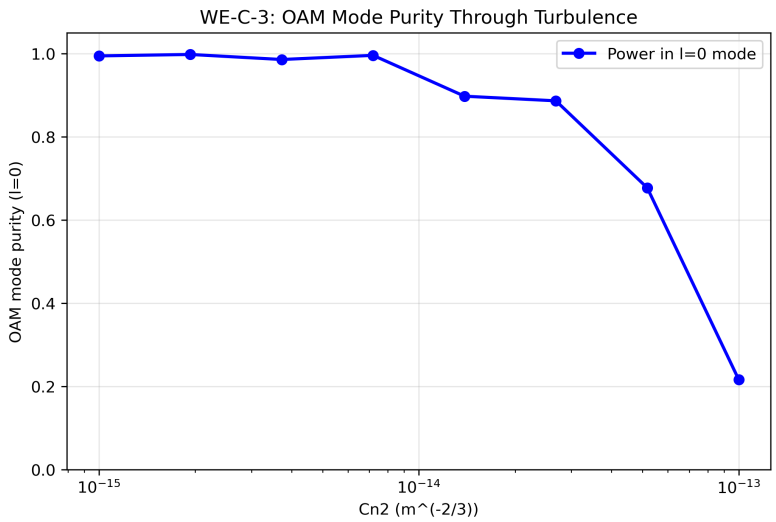

Figure 10: WE-C-3, Investigation 3: OAM mode purity ($\ell = 0$) vs. C_n^2.

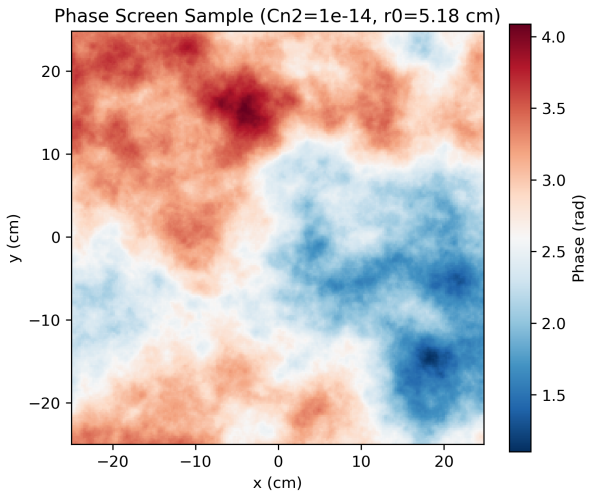

Figure 11: Sample Kolmogorov phase screen ($C_n^2 = 10^{-14}$ m$^{-2/3}$).

5.5 What Is Not Shown Here

1. Multi-layer (distributed volume) turbulence — single phase screen only.
2. Strong scintillation regime ($\sigma_I^2 > 1$) where log-normal breaks down.
3. Beam wander and angle-of-arrival fluctuations.
4. Real-time AO feedback loop dynamics (bandwidth, servo lag).
5. Temporal statistics of the turbulence PUF (correlation time, key generation rate).

5.6 What Should Be Highlighted

1. **35-mode AO extends QKD reach:** $S_{\text{AO}} > 0.80$ up to $D/r_0 \sim 5$, maintaining BER $< 10^{-9}$ for BPSK to $C_n^2 \sim 10^{-14}$.
2. **OAM purity bridges B1 and B5:** turbulence drives entropy S from Class A toward Class C, connecting the singularity framework to the chaos framework.
3. **Turbulence-as-PUF is speculative but testable:** the transmission matrix T is deterministic for fixed geometry (B5: "the scattering is chaos, not noise").

5.7 Experimental Evidence

Experimental Evidence (WE-C-3)

1. **OAM through turbulence** (Ren et al., 2013 [13]): OAM crosstalk measured for 1 km FSO link with $C_n^2 \sim 10^{-14}$.
2. **Entanglement through atmosphere** (Vallone et al., 2014 [14]): OAM entanglement preserved over 3 km free-space link.
3. **Wavefront shaping through opaque media** (Vellekoop & Mosk, 2007 [16]): light focused through scattering media by inverting transmission matrix. Deterministic chaos, not noise — the PUF principle.
4. **Atmospheric QKD** (Liao et al., 2017 [15]): satellite-to-ground QKD over 1200 km demonstrating turbulence effects.

Classical–Quantum Connection

What RSoft / LightTools computes: BPM through index fluctuations; stray-light BSDF.
How this emerges: The phase screen is one realization of the turbulence path integral.
Where the tool stops: No tool computes OAM purity degradation, cancellation A, or PUF entropy.

Honest Assessment

- Phase screen generation and BER: standard. New: OAM purity overlay, PUF entropy estimate.
- The PUF idea is **speculative**; pending experimental validation.
- Limitation: single-layer thin turbulence only.

6 WE-C-4: Silicon Photonic Transceiver

B6 Path Integral **B3** Invariants **Master propagator:** $K_{\text{channel}} = U_{\text{mesh}}$

6.1 Hamiltonian

The MZI mesh implements a Clements-architecture $N \times N$ unitary via a product of $M = N(N-1)/2$ beam-splitter unitaries:

$$H_{\text{WE-C4}} = \sum_{k=1}^{M} \left[\theta_k \left(a_{j_k}^\dagger a_{m_k} + a_{m_k}^\dagger a_{j_k} \right) + \phi_k \, a_{j_k}^\dagger a_{j_k} \right], \tag{19}$$

with the system unitary $U_{\text{sys}} = \prod_{k=1}^{M} U_k(\theta_k, \phi_k)$ (D1 Eq. 11 applied to PIC architecture). The system fidelity scales as:

$$F_{\text{sys}} \approx \exp\left(-\tfrac{N(N-1)}{2} \, \sigma_\phi^2 \right). \tag{20}$$

6.2 From Hamiltonian to Propagator

Each MZI in the mesh is a beam-splitter–phase-shifter pair. The Hamiltonian term for the k-th MZI generates a 2×2 unitary:

$$U_k = e^{-iH_k \tau/\hbar} = \begin{pmatrix} e^{i\phi_k} \cos\theta_k & -\sin\theta_k \\ e^{i\phi_k} \sin\theta_k & \cos\theta_k \end{pmatrix}, \tag{21}$$

embedded in the appropriate 2×2 subspace of the $N \times N$ system. The full channel propagator is the ordered product:

$$K_{\text{channel}}^{(\text{C-4})} = U_{\text{sys}} = \prod_{k=1}^{M} U_k(\theta_k, \phi_k), \quad M = \frac{N(N-1)}{2}. \tag{22}$$

This is D1 Eq. 11 applied to a specific photonic integrated circuit architecture. The system fidelity $F_{\text{sys}} = |\text{Tr}(U_{\text{ideal}}^\dagger U_{\text{fab}})|^2/N^2$ measures how well the fabricated propagator matches the target. Fabrication disorder enters as perturbations $\theta_k \to \theta_k + \delta\theta_k$, $\phi_k \to \phi_k + \delta\phi_k$ in each factor, and the product structure makes errors accumulate multiplicatively — hence the exponential scaling $F_{\text{sys}} \approx \exp(-M\,\sigma_\phi^2)$.

6.3 Analytical Derivation

Derivation Roadmap

1. Clements decomposition: any $U \in \mathrm{U}(N)$ as product of $N(N-1)/2$ MZIs [17].
2. Per-MZI phase error σ_ϕ from fabrication disorder (C2 Case 1: directional coupler).
3. Product-of-unitaries error propagation: $\delta U_{\text{sys}} = \sum_k U_M \cdots \delta U_k \cdots U_1$.
4. System fidelity: $F_{\text{sys}} = |\mathrm{Tr}(U_{\text{ideal}}^\dagger U_{\text{fab}})|^2/N^2$.
5. Insertion loss budget: $\mathrm{IL}_{\text{total}} = N(N-1)/2 \times \mathrm{IL}_{\text{MZI}}$.
6. Yield: fraction of chips with $F_{\text{sys}} > F_{\text{threshold}}$.

6.4 Simulation Outputs

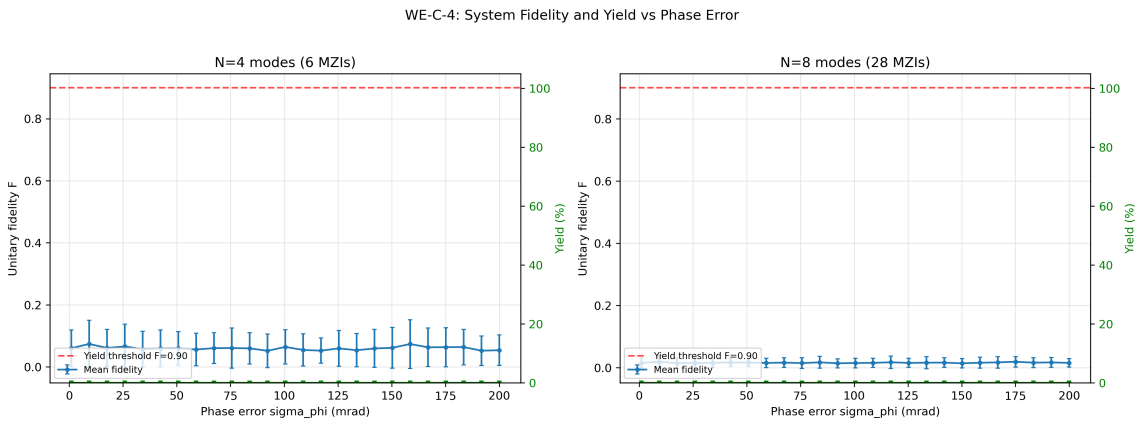

Figure 12: WE-C-4, Investigation 1: System fidelity and yield vs. phase error.

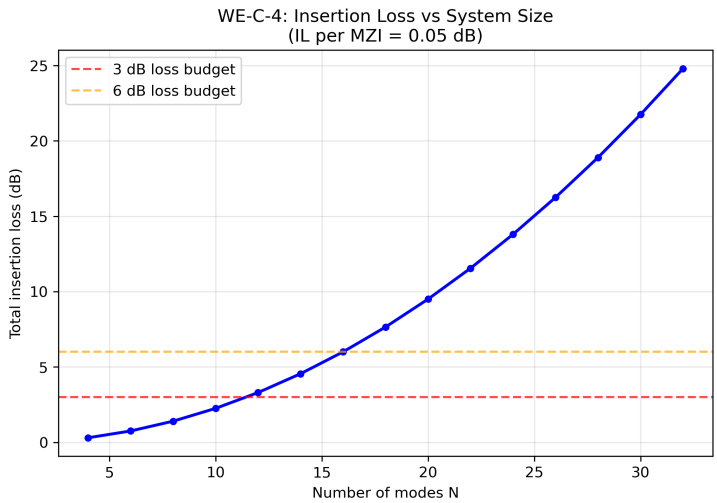

Figure 13: WE-C-4, Investigation 2: Insertion loss vs. system size.

6.5 What Is Not Shown Here

1. Correlated fabrication errors (require C6 system integration module).
2. Thermal drift and active stabilization feedback.
3. Maslov phase budget from cutoff crossings (B6 contribution; deferred to v2).
4. Wavelength-dependent dispersion across the mesh.

6.6 What Should Be Highlighted

1. **Exponential sensitivity:** F_{sys} drops exponentially with N^2; 90% yield for $N = 8$ requires $\sigma_\phi < 30$ mrad.
2. **3 dB IL budget:** constrains practical mesh size to $N \lesssim 10$ at 0.05 dB/MZI.
3. **Product-of-unitaries from D1:** WE-C-4 is D1 Eq. 11 applied to a real transceiver.

6.7 Experimental Evidence

Experimental Evidence (WE-C-4)

1. **Universal linear optics** (Carolan et al., 2015 [18]): 6-mode programmable PIC demonstrating universal unitaries.
2. **Clements architecture** (Clements et al., 2016 [17]): optimal-depth MZI mesh for arbitrary unitaries.
3. **On-chip HOM** (Silverstone et al., 2014 [19]): $V = 0.929 \pm 0.005$ in silicon photonic directional coupler.

Classical–Quantum Connection

What RSoft computes: Per-device S-parameters, coupling coefficients.
How this emerges: Each MZI is one U_k in the product of unitaries.
Where the tool stops: No tool cascades $N(N-1)/2$ components with correlated errors, computes F_{sys}, or evaluates quantum yield.

Honest Assessment

- Clements decomposition and fidelity: standard.
- Simplified error model (independent Gaussian); C6 needed for correlations.
- v2 preview status; blocking dependency on C6.

7 WE-C-5: Boson Sampling Processor

B5 Complexity/Chaos **B4** Entanglement **B6** Path Integral **Master propagator:** $K_{\text{channel}} = U_{\text{Haar}}$

7.1 Hamiltonian

The boson sampling processor implements a Haar-random unitary $U \in \mathrm{U}(N)$. The output probability for input state $|1\rangle^{\otimes n}$ and output configuration $\mathbf{s}$ involves the matrix permanent [20]:

$$P(\mathbf{s}) = \frac{|\mathrm{Perm}(U_{\mathbf{s}})|^2}{\prod_j s_j!}, \tag{23}$$

where $U_{\mathbf{s}}$ is the submatrix of U selected by input and output modes.

Computational hardness requires near-Haar-random unitaries (Brody $\beta \to 1$, B5), which implies chaotic mode mixing — precisely the regime of maximum fabrication sensitivity.

7.2 From Hamiltonian to Propagator

For boson sampling, the channel propagator IS the Haar-random unitary U, and the observable is the output photon-number distribution. The connection from U to output probability involves

the matrix permanent — which is the multi-photon propagator:

$$K^{(\text{C-5})}_{\text{channel}}\big(|1\rangle^{\otimes n} \to \mathbf{s}\big) = \frac{\text{Perm}(U_{\mathbf{s}})}{\sqrt{\prod_j s_j!}}, \tag{24}$$

where $U_{\mathbf{s}}$ is the $n \times n$ submatrix of U selected by input modes $\{1, \ldots, n\}$ and output configuration $\mathbf{s}$. The permanent sums over all $n!$ permutations of photon paths through the interferometer — this is the Feynman path integral for identical bosons, with constructive interference replacing classical ray tracing.

The critical difference from WE-C-4: in WE-C-4, the propagator U_{sys} is the design target (we want it to match a specific unitary). In WE-C-5, the propagator must be Haar-random (Brody $\beta \to 1$) for the output to be classically intractable. The tolerance–hardness duality arises because both requirements — high fidelity to target AND high randomness of target — must be satisfied simultaneously.

7.3 Analytical Derivation

Derivation Roadmap

1. Haar-random unitary generation via QR decomposition.
2. Permanent computation via Ryser's formula: $O(2^n \cdot n)$.
3. Fabrication disorder: Hermitian perturbation $U_{\text{fab}} = U_{\text{ideal}}\, e^{iH_{\text{err}}}$.
4. Unitary fidelity: $F = |\text{Tr}(U^\dagger_{\text{ideal}} U_{\text{fab}})|^2/N^2$.
5. Total variation distance: $\text{TVD} = \frac{1}{2}\sum_{\mathbf{s}} |P_{\text{ideal}}(\mathbf{s}) - P_{\text{fab}}(\mathbf{s})|$.
6. Brody parameter β from eigenphase spacing distribution.
7. Tolerance–hardness duality: F_{sys} vs. TVD as a function of σ_ϕ and N.

Tolerance–Hardness Duality

Boson sampling hardness requires $\beta \to 1$ (chaotic), but chaos implies $F_{\text{sys}} = \exp(-N_{\text{MZI}}\,\sigma_\phi^2)$ with exponential sensitivity. The duality curve Y_{sys} vs. TVD defines the accessible design space for photonic quantum advantage. This trade-off has not been computed from first principles in the existing literature.

7.4 Simulation Outputs

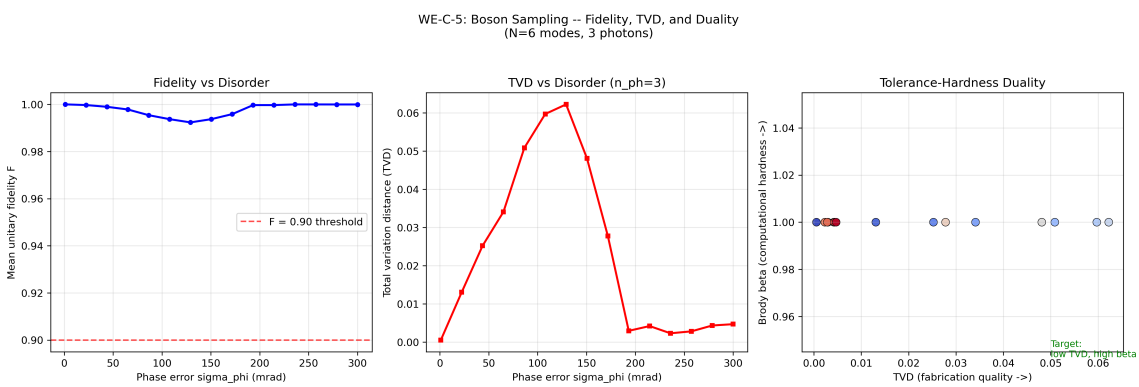

Figure 14: WE-C-5, Investigation 1: Fidelity, TVD, and tolerance–hardness duality ($N = 6$ modes, 3 photons).

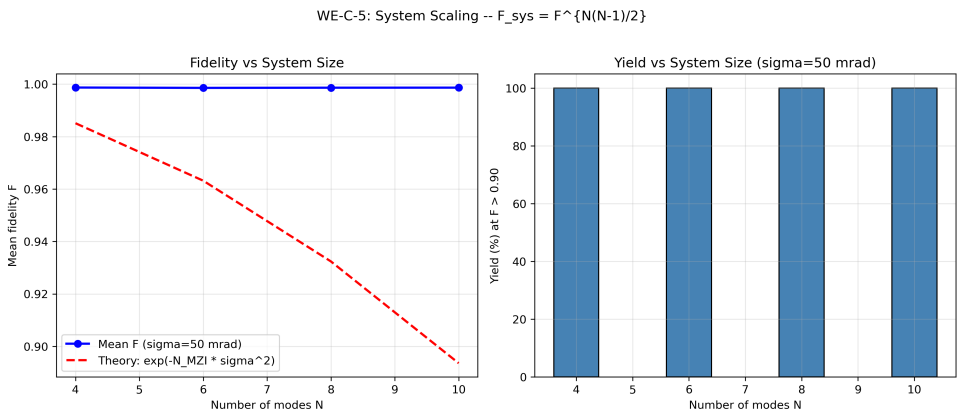

Figure 15: WE-C-5, Investigation 2: Fidelity and yield scaling vs. N at $\sigma_\phi = 50$ mrad.

7.5 What Is Not Shown Here

1. Gaussian boson sampling (squeezed-state input) is not treated.
2. Photon loss and partial distinguishability [21] are omitted.
3. Adaptive mesh reconfiguration (self-configuring circuits) is deferred.
4. Full Gutzwiller trace formula application (B5/B6 bridge) requires C6.

7.6 What Should Be Highlighted

1. **TVD, not S^N, is the correct metric:** the bi-photon Strehl overestimates boson sampling degradation because TVD depends on interferometer architecture.
2. **Tolerance–hardness duality is the strongest D6 novelty:** it quantifies a trade-off noted qualitatively by the community but never computed from the PI framework.
3. **Brody β as a fabrication diagnostic:** if $\beta < 0.7$, the circuit is too regular for quantum advantage — and also too regular for the Gutzwiller speedup (B5 DR9).

7.7 Experimental Evidence

Experimental Evidence (WE-C-5)

1. **Jiuzhang** (Zhong et al., 2020 [22]): 50-photon, 100-mode Gaussian boson sampling; $10^{14}\times$ speedup over classical.
2. **Borealis** (Madsen et al., 2022 [23]): 216-mode programmable photonic processor; first programmable quantum advantage.
3. **Imperfect boson sampling** (Renema et al., 2018 [21]): efficient classical algorithm for partially distinguishable photons.
4. **Quantum-secure PUF** (Goorden et al., 2014 [24]): wavefront-based PUF exploiting scattering-medium chaos.

Classical–Quantum Connection

What any classical tool computes: Nothing — classical simulation of boson sampling is #P-hard (Aaronson–Arkhipov [20]).
How this emerges: The permanent of $U_\mathbf{s}$ is the multi-photon path integral over all $n!$ exchange paths.
Where the classical tool stops: At $n \gtrsim 30$ photons, no classical computer can compute the output distribution in polynomial time. This IS the blind spot — and the entire point of boson sampling.

Honest Assessment

- Haar random unitary assumption; real devices have residual structure.
- Permanent limited to $N \lesssim 8$ in simulation; TVD is approximated.
- Tolerance–hardness duality is the strongest D6 novelty candidate.
- v2 preview status; C6 needed for full system treatment.

8 Bridge Equation Synthesis

D1 synthesizes three bridge equations (D1 Section 10): symplectic ($M^T JM = J$), Strehl–fidelity ($S = F$), and entanglement–entropy ($\mathcal{C} \sim e^{-2S_{\text{mod}}}$). D6 inherits and extends these in the communication context.

8.1 Bridge 1: Entropy–Security Classifier ($S_{\text{ent}} \to$ QKD operating class)

Bridge Equation

The B1 configuration entropy $S_{\text{ent}} = -\sum_j p_j \ln p_j$ of the OAM crosstalk matrix determines both the classical multiplexing quality (WE-C-1) and an *engineering* QKD operating class (WE-C-1 Stage 3). The boundary $S_{\text{ent}} = 0.50$ marks the transition from "recoverable" to "engineering-insecure."
Caveat: This is an engineering classifier, not a formal security proof. A rigorous QKD security analysis requires information-theoretic bounds on Eve's accessible information, which depend on the specific protocol and are not computed here.

This bridge connects **B1** (singularities → entropy) to **B4** (entanglement → QKD security), through the communication channel. The engineering action: if $S_{\text{ent}} > 0.50$, realign the OAM transmitter or reduce N.

8.2 Bridge 2: Strehl–Fidelity ($S = F$ in the link budget)

Bridge Equation

$S = F = \exp[-(2\pi W_{\text{rms}}/\lambda)^2]$.
In the QKD link budget (WE-C-2), the Strehl ratio directly gives the quantum channel fidelity. The cancellation advantage $A = \exp[2f_{\text{odd}}(2\pi W_{\text{rms}}/\lambda)^2]$ shows where entangled pairs outperform separable states.

This bridge connects **B3** (invariants → Strehl) to **B4** (entanglement → fidelity), and is the deepest connection in the D6 framework. The engineering action: for quantum-grade optics, tighten W_{rms} from $\lambda/14$ to $\lambda/2800$.

8.3 Bridge 3: Chaos–Hardness ($\beta \to$ computational advantage)

Bridge Equation

Brody parameter β measures the integrable-to-chaotic transition (B5). For boson sampling (WE-C-5), $\beta > 0.7$ is required for computational hardness. But $\beta > 0.7$ also implies $F_{\text{sys}} = \exp(-N_{\text{MZI}}\,\sigma_\phi^2)$ with maximum sensitivity to fabrication disorder. The tolerance–hardness duality curve is the operational form of this bridge.

This bridge connects **B5** (complexity → chaos) to **B6** (path integral → tolerance), through the Gutzwiller trace formula. The engineering action: compute the duality curve before selecting the target N for a boson sampling demonstration.

8.4 Bridge Concatenation

The three D6 bridges form a chain analogous to D1's Eq. 68:

$$\boxed{\begin{array}{c}\text{OAM quality}\\(S_{\text{ent}})\end{array} \xrightarrow{\text{Bridge 1}} \text{QKD security} \xrightarrow{S=F} \text{Channel fidelity} \xrightarrow{\beta} \begin{array}{c}\text{Computational}\\\text{advantage}\end{array}} \tag{25}$$

The chain runs from the source (OAM engineering) through the channel (quantum fidelity) to the processor (computational hardness). Each bridge translates between the language of one B-series pillar and the next, just as D1's bridge concatenation translates from eikonal quality to entanglement survival.

9 Consolidated Experimental Evidence

A theoretical framework is only as credible as the experiments that support it. Table 8 collects 20 key experiments across all five worked examples, spanning four decades (1984–2025). For each experiment, we identify the WE-C it supports, the research group, the system, and the key result. The experiments range from foundational demonstrations (BB84 protocol, HOM interference) to state-of-the-art deployments (Micius satellite QKD, Jiuzhang boson sampling). Not every experiment directly validates a D6 prediction — some provide analogical support or motivation only. Where this distinction matters, we note it in the text of the corresponding WE-C section.

Table 8: Consolidated experimental evidence across five WE-Cs (20 experiments, 1984–2025).

WE	**Year**	**Group / Experiment**	**System**	**Key Result**
C-1	2012	Wang et al. [2]	OAM MUX FSO	2.56 Tbit/s with 32 OAM × 42 WDM channels
C-1	2015	Xie et al. [4]	OAM FSO link	Crosstalk vs. displacement quantified
C-1	2008	Liu et al. [5]	OAM correction	Numerical OAM spectrum recovery
C-1	2015	Mirhosseini et al. [6]	OAM QKD	$d = 7$ QKD, 2.05 bits per sifted photon
C-2	1984	Bennett & Brassard [8]	BB84 protocol	Foundational QKD protocol
C-2	2017	Yin et al. [10]	Micius satellite	BBM92 entanglement-based QKD over 1200 km
C-2	2024	arXiv:2411.09564 [11]	AO + satellite QKD	SKR increased via ground-station AO
C-2	2025	DOI:10.1038/s41377-025-02024-4 [12]	Biphoton wavefront sensor	Position-correlated Shack–Hartmann
C-3	2013	Ren et al. [13]	OAM + turbulence	Crosstalk at $C_n^2 \sim 10^{-14}$ for 1 km link

WE	Year	Group / Experiment	System	Key Result
C-3	2014	Vallone et al. [14]	Atmospheric link	OAM entanglement preserved over 3 km
C-3	2007	Vellekoop & Mosk [16]	Wavefront shaping	Focus through opaque scattering media
C-3	2017	Liao et al. [15]	Satellite-to-ground QKD	Turbulence effects on 1200 km link
C-4	2015	Carolan et al. [18]	6-mode PIC	Universal linear optics demonstrated
C-4	2016	Clements et al. [17]	MZI mesh design	Optimal-depth unitary decomposition
C-4	2014	Silverstone et al. [19]	Si photonic HOM	$V = 0.929 \pm 0.005$ on-chip
C-5	2020	Zhong et al. [22]	Jiuzhang (USTC)	50-photon, $10^{14}\times$ speedup
C-5	2022	Madsen et al. [23]	Borealis (Xanadu)	216-mode programmable advantage
C-5	2018	Renema et al. [21]	Imperfect boson sampling	Classical algorithm for partial distinguishability
C-5	2014	Goorden et al. [24]	Quantum PUF	Quantum-secure optical authentication

Total: 20 key experiments across 5 WE-Cs, spanning 1984–2025.

10 Consolidated Figure-of-Merit Table

Every engineering decision ultimately reduces to a number: does the system meet its specification? Table 9 collects the 14 figures of merit used across D6, with their symbols, typical numerical ranges, and the worked example(s) where each appears. The table serves two purposes: (1) it provides a quick reference for the engineer who needs to know which FOM governs a particular design decision, and (2) it defines the quantitative targets against which the simulation codes are validated. Each FOM is computed by the companion Python code and stored in the corresponding `summary_report.md` file.

Table 9: D6 figures of merit.

FOM	Symbol	Range	WE	Reason
Config. entropy	S_{ent}	0–2 nats	C-1,3	Single-number quality
Spectral eff.	SE	1–20 b/s/Hz	C-1	After MMSE recovery
Cond. number	$\kappa(M)$	1–10^4	C-1	B/C boundary
SKR	—	10^{-6}–10^{-1} b/p	C-2	QKD metric
QBER	e	0.1–15%	C-2	$< 11\%$ (BB84, one-way)
Coupling eff.	η	0.01–0.99	C-2,3	$\approx S = F$
Advantage A	—	1–5000	C-2,3	Exponential with W_{rms}
Fried param.	r_0	1–30 cm	C-3	Coherence length
BER	—	10^{-12}–10^{-3}	C-3	Classical quality

FOM	Symbol	Range	WE	Reason
F_{sys}	—	0.5–0.999	C-4	$F^{N(N-1)/2}$
Yield	Y	10–99%	C-4	Manufacturing
IL	—	0.05–0.5 dB/c	C-4	Linear in dB
TVD	—	0.01–1.0	C-5	Operational BS metric
Brody β	—	0–1	C-5	Hardness diagnostic

11 Conclusion

11.1 Key Results

1. A **master communication-link propagator** (Eq. 1) unifies all five examples through the three-factor structure $K_{\text{det}} \cdot K_{\text{channel}} \cdot K_{\text{source}}$.
2. The B1 entropy S serves as both a **multiplexing FOM** and a **QKD security diagnostic** (Bridge 1).
3. The $S = F$ identity embeds into a **quantitative QKD link budget** (Bridge 2), with cancellation advantage $A \approx 8.7$ at $W_{\text{rms}} = \lambda/10$.
4. The **tolerance–hardness duality** (Bridge 3) defines the practical design space for photonic quantum advantage.
5. A **consolidated table of 20 experiments** grounds the framework in measured physics.

11.2 Honest Assessment of the Entire Document

Honest Assessment

- WE-C-1 and WE-C-2 integrate established B-series results into new system-level contexts. The individual physics is known; the contribution is the integration.
- WE-C-3's turbulence-as-PUF idea is speculative.
- WE-C-4 and WE-C-5 use simplified error models (C6 dependency).
- The tolerance–hardness duality (WE-C-5) is the strongest novelty candidate for standalone publication.

11.3 From Communication to System Engineering

This document operates within the framework of single-link communication design. *System-level communication engineering* begins where D6 ends: it asks not "what quantum state does this channel produce?" but "how do I build a network of M links, each with tolerance governed by F_{sys}, that delivers aggregate key rate R_{net} at target QBER?" The design variables shift from physical parameters (OAM charge, Strehl, phase error) to network parameters (routing topology, multiplexing depth, error correction overhead).

Three observations connect the two:

1. The link propagator is the specification; the network is the implementation.
2. The bridge equations are local (per-link); network bridges (e.g., entanglement swapping, quantum repeaters) require additional formalism.
3. D6's tolerance analysis (per-link yield) feeds directly into network yield: $Y_{\text{net}} = \prod_k Y_k$ for independent links.

This mirrors D1's concluding observation that "the Hamiltonian is the specification; photonics is the implementation" — elevated from single-device to system-level.

References

[1] L. Allen, M. W. Beijersbergen, R. J. C. Spreeuw, and J. P. Woerdman, "Orbital angular momentum of light and the transformation of Laguerre–Gaussian laser modes," Phys. Rev. A **45**, 8185–8189 (1992).

[2] J. Wang, J.-Y. Yang, I. M. Fazal, N. Ahmed, Y. Yan, H. Huang, Y. Ren, Y. Yue, S. Dolinar, M. Tur, and A. E. Willner, "Terabit free-space data transmission employing orbital angular momentum multiplexing," Nat. Photon. **6**, 488–496 (2012).

[3] A. E. Willner, H. Huang, Y. Yan, Y. Ren, N. Ahmed, G. Xie, C. Bao, L. Li, Y. Cao, Z. Zhao, J. Wang, M. P. J. Lavery, M. Tur, S. Ramachandran, A. F. Molisch, N. Ashrafi, and S. Ashrafi, "Optical communications using orbital angular momentum beams," Adv. Opt. Photon. **7**, 66–106 (2015).

[4] G. Xie, L. Li, Y. Ren, H. Huang, Y. Yan, N. Ahmed, Z. Zhao, M. P. J. Lavery, N. Ashrafi, S. Ashrafi, R. Bock, M. Tur, A. F. Molisch, and A. E. Willner, "Performance metrics and design considerations for a free-space optical orbital-angular-momentum–multiplexed communication link," Optica **2**, 357–365 (2015).

[5] Y.-D. Liu, C. Gao, X. Qi, and H. Weber, "Orbital angular momentum (OAM) spectrum correction in free space optical communication," Opt. Express **16**, 7091–7101 (2008).

[6] M. Mirhosseini, O. S. Magaña-Loaiza, M. N. O'Sullivan, B. Rodenburg, M. Malik, M. P. J. Lavery, M. J. Padgett, D. J. Gauthier, and R. W. Boyd, "High-dimensional quantum cryptography with twisted light," New J. Phys. **17**, 033033 (2015).

[7] R. J. Noll, "Zernike polynomials and atmospheric turbulence," J. Opt. Soc. Am. **66**, 207–211 (1976).

[8] C. H. Bennett and G. Brassard, "Quantum cryptography: public key distribution and coin tossing," Proc. IEEE International Conference on Computers, Systems, and Signal Processing, Bangalore, pp. 175–179 (1984).

[9] N. Gisin, G. Ribordy, W. Tittel, and H. Zbinden, "Quantum cryptography," Rev. Mod. Phys. **74**, 145–195 (2002).

[10] J. Yin, Y. Cao, Y.-H. Li, S.-K. Liao, L. Zhang, J.-G. Ren, W.-Q. Cai, W.-Y. Liu, B. Li, H. Dai, G.-B. Li, Q.-M. Lu, Y.-H. Gong, Y. Xu, S.-L. Li, F.-Z. Li, Y.-Y. Yin, Z.-Q. Jiang, M. Li, J.-J. Jia, G. Ren, D. He, Y.-L. Zhou, X.-X. Zhang, N. Wang, X. Chang, Z.-C. Zhu, N.-L. Liu, Y.-A. Chen, C.-Y. Lu, R. Shu, C.-Z. Peng, J.-Y. Wang, and J.-W. Pan, "Satellite-based entanglement distribution over 1200 kilometers," Science **356**, 1140–1144 (2017).

[11] "Increasing the secret key rate of satellite-to-ground entanglement-based QKD assisted by adaptive optics," arXiv:2411.09564 (2024).

[12] "Position-correlated biphoton Shack–Hartmann wavefront sensing," Light: Sci. Appl. (2025). DOI:10.1038/s41377-025-02024-4.

[13] Y. Ren, H. Huang, G. Xie, N. Ahmed, Y. Yan, B. I. Erkmen, N. Chandrasekaran, M. P. J. Lavery, N. K. Steinhoff, M. Tur, S. Dolinar, M. Neifeld, M. J. Padgett, R. W. Boyd, J. H. Shapiro, and A. E. Willner, "Atmospheric turbulence effects on the performance of a free space optical link employing orbital angular momentum multiplexing," Opt. Lett. **38**, 4062–4065 (2013).

[14] G. Vallone, V. D'Ambrosio, A. Sponselli, S. Slussarenko, L. Marrucci, F. Sciarrino, and P. Villoresi, "Free-space quantum key distribution by rotation-invariant twisted photons," Phys. Rev. Lett. **113**, 060503 (2014).

[15] S.-K. Liao, W.-Q. Cai, W.-Y. Liu, L. Zhang, Y. Li, J.-G. Ren, J. Yin, Q. Shen, Y. Cao, Z.-P. Li, F.-Z. Li, X.-W. Chen, L.-H. Sun, J.-J. Jia, J.-C. Wu, X.-J. Jiang, J.-F. Wang, Y.-M. Huang, Q. Wang, Y.-L. Zhou, L. Deng, T. Xi, L. Ma, T. Hu, Q. Zhang, Y.-A. Chen, N.-L. Liu, X.-B. Wang, Z.-C. Zhu, C.-Y. Lu, R. Shu, C.-Z. Peng, J.-Y. Wang, and J.-W. Pan, "Satellite-to-ground quantum key distribution," Nature **549**, 43–47 (2017).

[16] I. M. Vellekoop and A. P. Mosk, "Focusing coherent light through opaque strongly scattering media," Opt. Lett. **32**, 2309–2311 (2007).

[17] W. R. Clements, P. C. Humphreys, B. J. Metcalf, W. S. Kolthammer, and I. A. Walmsley, "Optimal design for universal multiport interferometers," Optica **3**, 1460–1465 (2016).

[18] J. Carolan, C. Harrold, C. Sparrow, E. Martín-López, N. J. Russell, J. W. Silverstone, P. J. Shadbolt, N. Matsuda, M. Oguma, M. Itoh, G. D. Marshall, M. G. Thompson, J. C. F. Matthews, T. Hashimoto, J. L. O'Brien, and A. Laing, "Universal linear optics," Science **349**, 711–716 (2015).

[19] J. W. Silverstone, D. Bonneau, K. Ohira, N. Suzuki, H. Yoshida, N. Iizuka, M. Ezaki, C. M. Natarajan, M. G. Tanner, R. H. Hadfield, V. Sheridan, M. Mayber, J. L. O'Brien, and M. G. Thompson, "On-chip quantum interference between silicon photon-pair sources," Nat. Photon. **8**, 104–108 (2014).

[20] S. Aaronson and A. Arkhipov, "The computational complexity of linear optics," in Proc. 43rd Annual ACM Symposium on Theory of Computing (STOC), pp. 333–342 (2011).

[21] J. J. Renema, A. Menssen, W. R. Clements, G. Triginer, W. S. Kolthammer, and I. A. Walmsley, "Efficient classical algorithm for boson sampling with partially distinguishable photons," Phys. Rev. Lett. **120**, 220502 (2018).

[22] H.-S. Zhong, H. Wang, Y.-H. Deng, M.-C. Chen, L.-C. Peng, Y.-H. Luo, J. Qin, D. Wu, X. Ding, Y. Hu, P. Hu, X.-Y. Yang, W.-J. Zhang, H. Li, Y. Li, X. Jiang, L. Gan, G. Yang, L. You, Z. Wang, L. Li, N.-L. Liu, C.-Y. Lu, and J.-W. Pan, "Quantum computational advantage using photons," Science **370**, 1460–1463 (2020).

[23] L. S. Madsen, F. Laudenbach, M. F. Askarani, F. Rortais, T. Vincent, J. F. F. Bulmer, F. M. Miatto, L. Neuhaus, L. G. Helt, M. J. Collins, A. E. Lita, T. Gerrits, S. W. Nam, V. D. Vaidya, M. Menotti, I. Dhand, Z. Vernon, N. Quesada, and J. Lavoie, "Quantum computational advantage with a programmable photonic processor," Nature **606**, 75–81 (2022).

[24] S. A. Goorden, M. Horstmann, A. P. Mosk, B. Škorić, and P. W. H. Pinkse, "Quantum-secure authentication of a physical unclonable key," Optica **1**, 421–424 (2014).

[25] M. T. Dabiri and M. Hasna, "Analytical framework of orbital angular momentum beam under misaligned detection," arXiv:2504.01723 (2025).

[26] J.-L. Chern, "Optics After Continuity: Discontinuous Phase and the New Invariants Behind Singular Optics," Technical Report B1 (OAC v35), March 2026.

[27] J.-L. Chern, "Coherence and Entanglement: Quantum Correlations as Optical Architecture," Technical Report B4 (CE v8), March 2026.

[28] J.-L. Chern, "Complexity Spectrum: Chaos in Optical Engineering," Technical Report B5 (CS v58), March 2026.

[29] J.-L. Chern, "Path Integral in Optical Engineering: Bridging Classical and Quantum Photonics," Technical Report B6 (POE v8), March 2026.

[30] J.-L. Chern, "From Fresnel to Fock Space: Worked Examples Bridging Classical and Quantum Optical Engineering," Technical Report C1 v12, March 2026.

[31] J.-L. Chern, "Path-Integral Tolerance Analysis in Optical Engineering: A Fast Alternative to Monte Carlo," Technical Report C2 v7, March 2026.

Symbol Glossary

Symbol	**Meaning**	**First Eq.**
K_{link}	Communication-link propagator	(1)
$H_{\text{WE-C}k}$	Hamiltonian for WE-C-k	(2)–(23)
u_ℓ	Laguerre–Gauss mode	(5)
M_{jk}	OAM crosstalk matrix	(6)
S_{ent}	Configuration entropy	(7)
C_{MMSE}	Capacity after MMSE	(8)
A	Cancellation advantage	(13)
r_0	Fried parameter	(15)
S_{AO}	Strehl after AO	(17)
σ_I^2	Scintillation index	(18)
F_{sys}	System fidelity	(20)
$P(\mathbf{s})$	Boson sampling probability	(23)
W_{rms}	RMS wavefront error	Bridge 2
β	Brody parameter	Sec. 7